

PRACTICAL-2

AIM: Hadoop Installation, Configuration and HDFS Operations on Windows/Linux

PROBLEM DEFINITION:

You have recently joined DataAnalytics Inc. as a Hadoop Administrator. The organization processes large-scale customer, sales, and marketing data using the Hadoop ecosystem. Due to increasing data volumes, the company has decided to migrate its data storage infrastructure from traditional file systems to Hadoop Distributed File System (HDFS).

As a Hadoop Administrator, your responsibility is to install and configure Hadoop in a SingleNode Cluster environment on Windows/Linux, verify the Hadoop services, manage HDFS storage, and perform data management operations.

Management wants a fully functional Hadoop environment that can store, retrieve, and process big data efficiently.

Your task is to install Hadoop, configure the Hadoop ecosystem, start Hadoop services, and perform HDFS operations using Hadoop commands.

Task:

- Hadoop Installation & Configuration ●
- Access Hadoop Web UI.

1. Create directories in HDFS.
2. Upload files to HDFS.
3. List HDFS files and directories.
4. Display file contents.
5. Copy files within HDFS.
6. Move files within HDFS.
7. Rename files.
8. Download files from HDFS.
9. Delete files and directories.
10. Check HDFS report.
11. Verify block information.
12. Check Hadoop cluster health.
13. Monitor Hadoop processes.
14. Check storage utilization.

SCREENSHOTS:

```

kali@kali: ~/Desktop/MarketingAnalytics
$ hdfs dfs -mkdir /sales
2026-07-17 05:58:25,412 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable

kali@kali: ~/Desktop/MarketingAnalytics
$ jps
3201 NameNode
3316 DataNode
4117 NodeManager
3526 SecondaryNameNode
3875 Jps
3995 ResourceManager

Startup Progress

kali@kali: ~/Desktop/MarketingAnalytics
$ hdfs dfs -ls /
2026-07-17 06:01:19,785 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Found 1 items
drwxr-xr-x - kali supergroup 0 2026-07-17 05:58 /sales
Completion 100%

kali@kali: ~/Desktop/MarketingAnalytics
$ hdfs dfs -mkdir /sales/data
2026-07-17 06:01:52,951 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Completion 100%

kali@kali: ~/Desktop/MarketingAnalytics
$ nano sales_data.txt
Completion 100%

kali@kali: ~/Desktop/MarketingAnalytics
$ hdfs dfs -put sales_data.txt /sales
Completion 100%

kali@kali: ~/Desktop/MarketingAnalytics
$ hdfs dfs -ls /sales
2026-07-17 06:06:46,403 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Found 2 items
drwxr-xr-x - kali supergroup 0 2026-07-17 06:01 /sales/data
-rw-r--r-- 1 kali supergroup 63 2026-07-17 06:06 /sales/sales_data.txt
Completion 100%

kali@kali: ~/Desktop/MarketingAnalytics
$ hdfs dfs -ls -R /
2026-07-17 06:08:12,684 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
drwxr-xr-x - kali supergroup 0 2026-07-17 06:06 /sales
drwxr-xr-x - kali supergroup 0 2026-07-17 06:01 /sales/data
-rw-r--r-- 1 kali supergroup 63 2026-07-17 06:06 /sales/sales_data.txt
Completion 100%

kali@kali: ~/Desktop/MarketingAnalytics
$ hdfs dfs -cat /sales/sales_data.txt
2026-07-17 06:08:39,526 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
ProductA 120
ProductB 80
ProductC 150
ProductA 200
ProductB 90
Completion 100%

kali@kali: ~/Desktop/MarketingAnalytics
$ hdfs dfs -cp /sales/sales_data.txt /sales/sales_copy.txt
2026-07-17 06:09:15,353 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable

kali@kali: ~/Desktop/MarketingAnalytics
$ hdfs dfs -ls /sales
2026-07-17 06:09:25,705 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Found 3 items
drwxr-xr-x - kali supergroup 0 2026-07-17 06:01 /sales/data
-rw-r--r-- 1 kali supergroup 63 2026-07-17 06:09 /sales/sales_copy.txt
-rw-r--r-- 1 kali supergroup 63 2026-07-17 06:06 /sales/sales_data.txt
Completion 100%

kali@kali: ~/Desktop/MarketingAnalytics
$ hdfs dfs -mkdir /backup
2026-07-17 06:10:18,452 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable

kali@kali: ~/Desktop/MarketingAnalytics
$ hdfs dfs -mv /sales/sales_copy.txt /backup
2026-07-17 06:10:26,743 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable

kali@kali: ~/Desktop/MarketingAnalytics
$ hdfs dfs -ls /backup
2026-07-17 06:10:31,839 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Found 1 items
-rw-r--r-- 1 kali supergroup 63 2026-07-17 06:09 /backup/sales_copy.txt
Completion 100%

kali@kali: ~/Desktop/MarketingAnalytics
$ hdfs dfs -mv /sales/sales_data.txt /sales/customer_sales.txt
2026-07-17 06:10:41,088 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable

kali@kali: ~/Desktop/MarketingAnalytics
$ hdfs dfs -ls /sales
2026-07-17 06:11:20,643 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Found 2 items
-rw-r--r-- 1 kali supergroup 63 2026-07-17 06:06 /sales/customer_sales.txt
drwxr-xr-x - kali supergroup 0 2026-07-17 06:01 /sales/data
Completion 100%

kali@kali: ~/Desktop/MarketingAnalytics
$ mkdir Downloads_HDFS
Completion 100%

kali@kali: ~/Desktop/MarketingAnalytics
$ hdfs dfs -get /sales/customer_sales.txt Downloads_HDFS/
2026-07-17 06:16:58,014 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable

kali@kali: ~/Desktop/MarketingAnalytics
$ ls Downloads_HDFS
customer_sales.txt
Completion 100%

kali@kali: ~/Desktop/MarketingAnalytics
$ hdfs dfs -rm /sales/customer_sales.txt
2026-07-17 06:18:02,806 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Deleted /sales/customer_sales.txt
Completion 100%

kali@kali: ~/Desktop/MarketingAnalytics
$ hdfs dfs -rmdir /sales
2026-07-17 06:18:24,675 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
rmdir: '/sales': Directory is not empty
Completion 100%

kali@kali: ~/Desktop/MarketingAnalytics
$ hdfs dfs -rm -x /backup
2026-07-17 06:18:43,560 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Deleted /backup
Completion 100%

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(kali@kali) [~/Desktop/MarketingAnalytics]
$ hdfs dfsadmin -report
2026-07-17 06:18:59,023 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Configured Capacity: 84053143552 (78.28 GB)
Present Capacity: 59472039788 (55.39 GB)
DFS Remaining: 59471982592 (55.39 GB)
DFS Used: 57196 (55.86 KB)
DFS Used%: 0.00%
Replicated Blocks:
  Under replicated blocks: 0
  Blocks with corrupt replicas: 0
  Missing blocks: 0
  Missing blocks (with replication factor 1): 0
  Low redundancy blocks with highest priority to recover: 0
  Pending deletion blocks: 0
Erasure Coded Block Groups:
  Low redundancy block groups: 0
  Block groups with corrupt internal blocks: 0
  Missing block groups: 0
  Low redundancy blocks with highest priority to recover: 0
  Pending deletion blocks: 0

Live datanodes (1):

Name: 127.0.0.1:9866 (localhost)
Hostname: kali
Decommission Status : Normal
Configured Capacity: 84053143552 (78.28 GB)
DFS Used: 57196 (55.86 KB)
Non DFS Used: 20264382612 (18.87 GB)
DFS Remaining: 59471982592 (55.39 GB)
DFS Used%: 0.00%
DFS Remaining%: 70.76%
Configured Cache Capacity: 0 (0 B)
Cache Used: 0 (0 B)
Cache Remaining: 0 (0 B)
Cache Used%: 100.00%
Cache Remaining%: 0.00%
Xceivers: 0
Last contact: Fri Jul 17 06:18:58 EDT 2026
Last Block Report: Fri Jul 17 04:52:25 EDT 2026
Num of Blocks: 0

```

```

(kali@kali) [~/Desktop/MarketingAnalytics]
$ hdfs fsck /
2026-07-17 06:19:16,296 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Connecting to namenode via http://localhost:9870/fsck?ugi=kali&path=%2F
FSCK started by kali (auth:SIMPLE) from /127.0.0.1 for path / at Fri Jul 17 06:19:17 EDT 2026

Status: HEALTHY
Number of data-nodes: 1
Number of racks: 1
Total dirs: 3
Total symlinks: 0

Replicated Blocks:
Total size: 0 B
Total files: 0
Total blocks (validated): 0
Minimally replicated blocks: 0
Over-replicated blocks: 0
Under-replicated blocks: 0
Mis-replicated blocks: 0
Default replication factor: 1
Average block replication: 0.0
Missing blocks: 0
Corrupt blocks: 0
Missing replicas: 0
Blocks queued for replication: 0

Erasure Coded Block Groups:
Total size: 0 B
Total files: 0
Total block groups (validated): 0
Minimally erasure-coded block groups: 0
Over-erasure-coded block groups: 0
Under-erasure-coded block groups: 0
Unsatisfactory placement block groups: 0
Average block group size: 0.0
Missing block groups: 0
Corrupt block groups: 0
Missing internal blocks: 0
Blocks queued for replication: 0
FSCK ended at Fri Jul 17 06:19:17 EDT 2026 in 25 milliseconds

```

```
(kali@kali)-[~/Desktop/MarketingAnalytics]
$ hdfs fsck /
2026-07-17 06:19:16,296 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Connecting to namenode via http://localhost:9870/fsc?ugi=kali&path=%2F
FSCK started by kali (auth:SIMPLE) from /127.0.0.1 for path / at Fri Jul 17 06:19:17 EDT 2026

Status: HEALTHY
Number of data-nodes: 1
Number of racks: 1
Total dirs: 3
Total symlinks: 0

Replicated Blocks:
Total size: 0 B
Total files: 0
Total blocks (validated): 0
Minimally replicated blocks: 0
Over-replicated blocks: 0
Under-replicated blocks: 0
Mis-replicated blocks: 0
Default replication factor: 1
Average block replication: 0.0
Missing blocks: 0
Corrupt blocks: 0
Missing replicas: 0
Blocks queued for replication: 0

Erasure Coded Block Groups:
Total size: 0 B
Total files: 0
Total block groups (validated): 0
Minimally erasure-coded block groups: 0
Over-erasure-coded block groups: 0
Under-erasure-coded block groups: 0
Unsatisfactory placement block groups: 0
Average block group size: 0.0
Missing block groups: 0
Corrupt block groups: 0
Missing internal blocks: 0
Blocks queued for replication: 0
FSCK ended at Fri Jul 17 06:19:17 EDT 2026 in 25 milliseconds

The filesystem under path '/' is HEALTHY
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```
(kali@kali)-[~/Desktop/MarketingAnalytics]
$ hdfs fsck / -files -blocks -locations
2026-07-17 06:19:35,256 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Connecting to namenode via http://localhost:9870/fsc?ugi=kali&files=1&blocks=1&locations=1&path=%2F
FSCK started by kali (auth:SIMPLE) from /127.0.0.1 for path / at Fri Jul 17 06:19:36 EDT 2026

/ <dir>
/sales <dir>
/sales/data <dir>

Status: HEALTHY
Number of data-nodes: 1
Number of racks: 1
Total dirs: 3
Total symlinks: 0

Replicated Blocks:
Total size: 0 B
Total files: 0
Total blocks (validated): 0
Minimally replicated blocks: 0
Over-replicated blocks: 0
Under-replicated blocks: 0
Mis-replicated blocks: 0
Default replication factor: 1
Average block replication: 0.0
Missing blocks: 0
Corrupt blocks: 0
Missing replicas: 0
Blocks queued for replication: 0

Erasure Coded Block Groups:
Total size: 0 B
Total files: 0
Total block groups (validated): 0
Minimally erasure-coded block groups: 0
Over-erasure-coded block groups: 0
Under-erasure-coded block groups: 0
Unsatisfactory placement block groups: 0
Average block group size: 0.0
Missing block groups: 0
Corrupt block groups: 0
Missing internal blocks: 0
Blocks queued for replication: 0
FSCK ended at Fri Jul 17 06:19:36 EDT 2026 in 2 milliseconds
```

```
The filesystem under path '/' is HEALTHY
```

```
Live datanodes (1):
Name: 127.0.0.1:9866 (localhost)
Hostname: kali
Decommission Status : Normal
Configured Capacity: 84053143552 (78.28 GB)
DFS Used: 57196 (55.86 KB)
Non DFS Used: 2964382612 (18.87 GB)
DFS Remaining: 59471982592 (55.39 GB)
DFS Used%: 0.00%
DFS Remaining%: 70.76%
Configured Cache Capacity: 0 (0 B)
Cache Used: 0 (0 B)
Cache Remaining: 0 (0 B)
Cache Used%: 100.00%
Cache Remaining%: 0.00%
Xceivers: 0
Last contact: Fri Jul 17 06:20:01 EDT 2026
Last Block Report: Fri Jul 17 04:52:25 EDT 2026
Num of Blocks: 0
```

```
[kali@kali]~/.Desktop/MarketingAnalytics$ hdfs dfs -du -h /
2026-07-17 06:20:51,446 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
0 0 /sales
```

KEY QUESTIONS / ANALYSIS:

1) Why is Hadoop preferred for Big Data storage?

Hadoop is preferred because it can store and process massive amounts of structured and unstructured data across multiple machines. It is scalable, cost-effective, fault-tolerant, and provides high availability through the Hadoop Distributed File System (HDFS).

2) What is the role of NameNode and DataNode in HDFS?

- NameNode: It is the master node that manages the HDFS namespace, stores metadata, and keeps track of file locations and permissions.
- DataNode: It is the worker node that stores the actual data blocks, serves read/write requests, and sends regular status reports (heartbeats) to the NameNode.

3) What is the difference between HDFS and traditional file systems?

HDFS	Traditional File System
Designed for Big Data storage	Designed for single-system storage
Stores data across multiple machines	Stores data on a single machine
Highly fault-tolerant through replication	Limited fault tolerance
Optimized for large files and sequential access	Suitable for small and medium-sized files
Easily scalable by adding new nodes	Scalability is limited

4) What are the advantages of distributed storage?

- Stores very large amounts of data across multiple machines.
- Provides fault tolerance through data replication.
- Improves performance by enabling parallel data access.
- Easily scalable by adding more nodes to the cluster.
- Ensures high availability even if one or more nodes fail.
- Reduces storage costs by using commodity hardware.
- Supports data recovery and disaster management through data replication.

- Allows multiple users and applications to access data simultaneously, improving efficiency.

CONCLUSION:

In this practical, Hadoop was successfully installed and configured in a single-node cluster environment. Various HDFS operations, including creating directories, uploading, downloading, copying, moving, renaming, listing, and deleting files, were performed successfully. The Hadoop Web UI was used to verify services, while cluster health, storage utilization, and running processes were monitored. This practical enhanced the understanding of Hadoop architecture and HDFS, demonstrating how distributed storage provides scalability, fault tolerance, reliability, and efficient management of large-scale data.